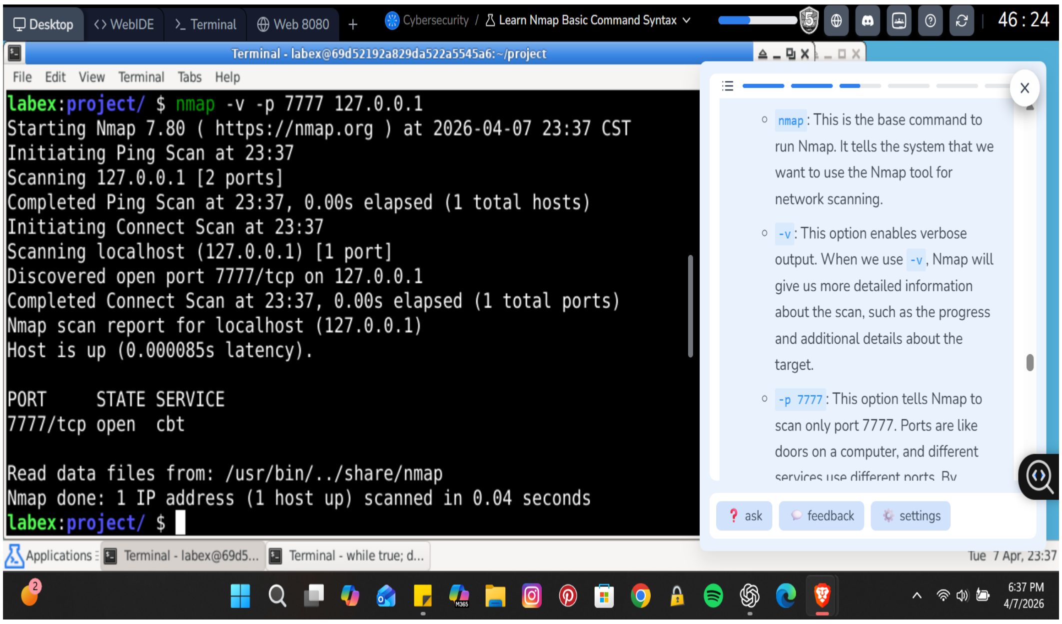
Nmap Project 3 – Basic Command Syntax & Network Scanning

This project demonstrates practical usage of Nmap for network reconnaissance and port scanning. The objective was to understand how to identify open ports, scan individual hosts, and perform network-wide scans in a controlled lab environment.

1. Localhost Port Scanning

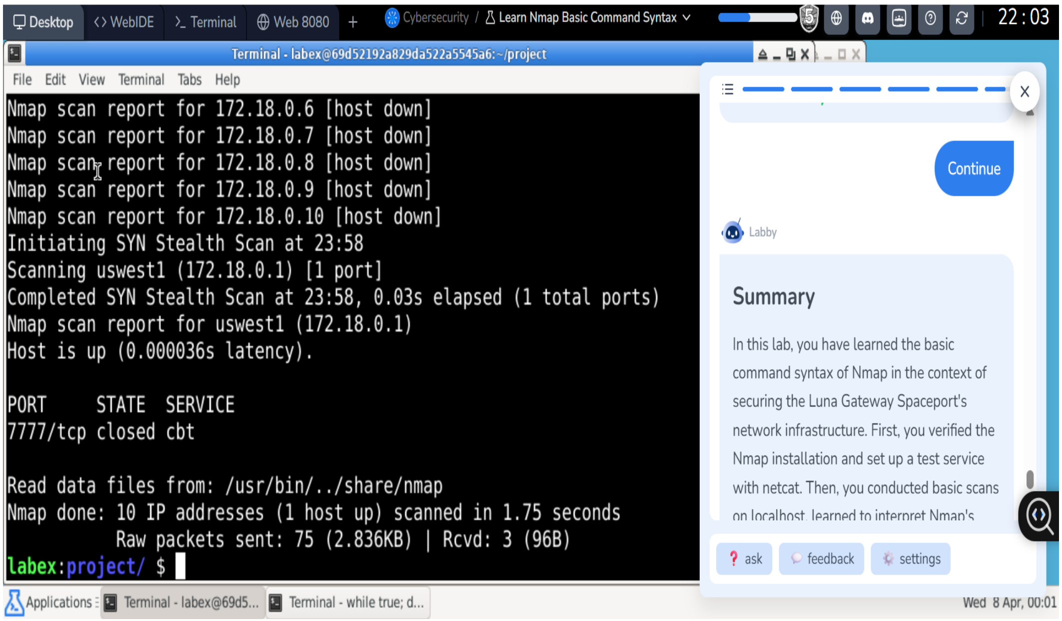
A test service was created using Netcat on port 7777. The following command was used: nmap -v -p 7777 127.0.0.1 This scan targeted the local machine (localhost) and checked whether port 7777 was open. The verbose (-v) flag provided detailed output during execution.



The results confirmed that port 7777/tcp was open, meaning the Netcat service was successfully running. This demonstrates how Nmap can detect active services on a host.

2. Network-Wide Scanning

A broader scan was conducted across multiple IP addresses within the same subnet using: sudo nmap -v -p 7777 172.18.0.1-10 > network_scan.txt This command scanned 10 hosts and saved the output to a file for analysis.



The scan results showed that most hosts were down, while one host (172.18.0.1) was active. However, port 7777 was closed on that host. This highlights how Nmap distinguishes between active hosts and available services.

Conclusion

This project demonstrates foundational Nmap skills, including: - Targeted port scanning - Interpreting scan results - Network-wide reconnaissance These techniques are essential in cybersecurity for identifying exposed services and potential vulnerabilities.